Software Requirements and Deployment

SOEN 342 - Section ?

The Project: Phase 1

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Fall 2026

Date date: Friday October 23rd, 2026

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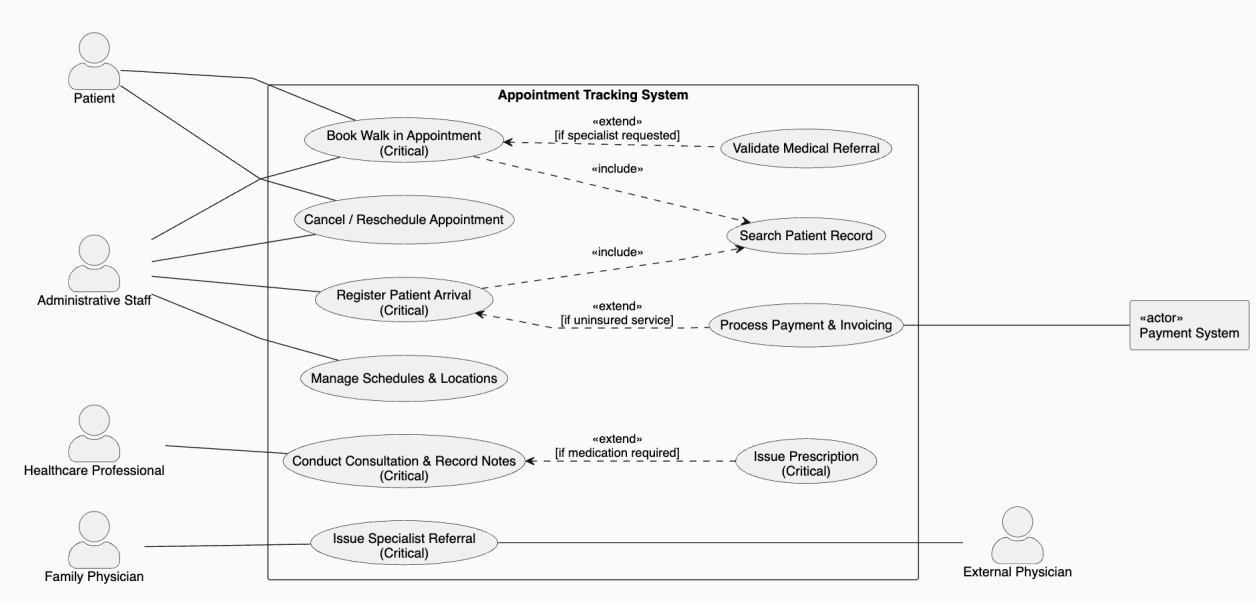
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I. Use Case Diagram (incl. crit./non-crit. Use cases)



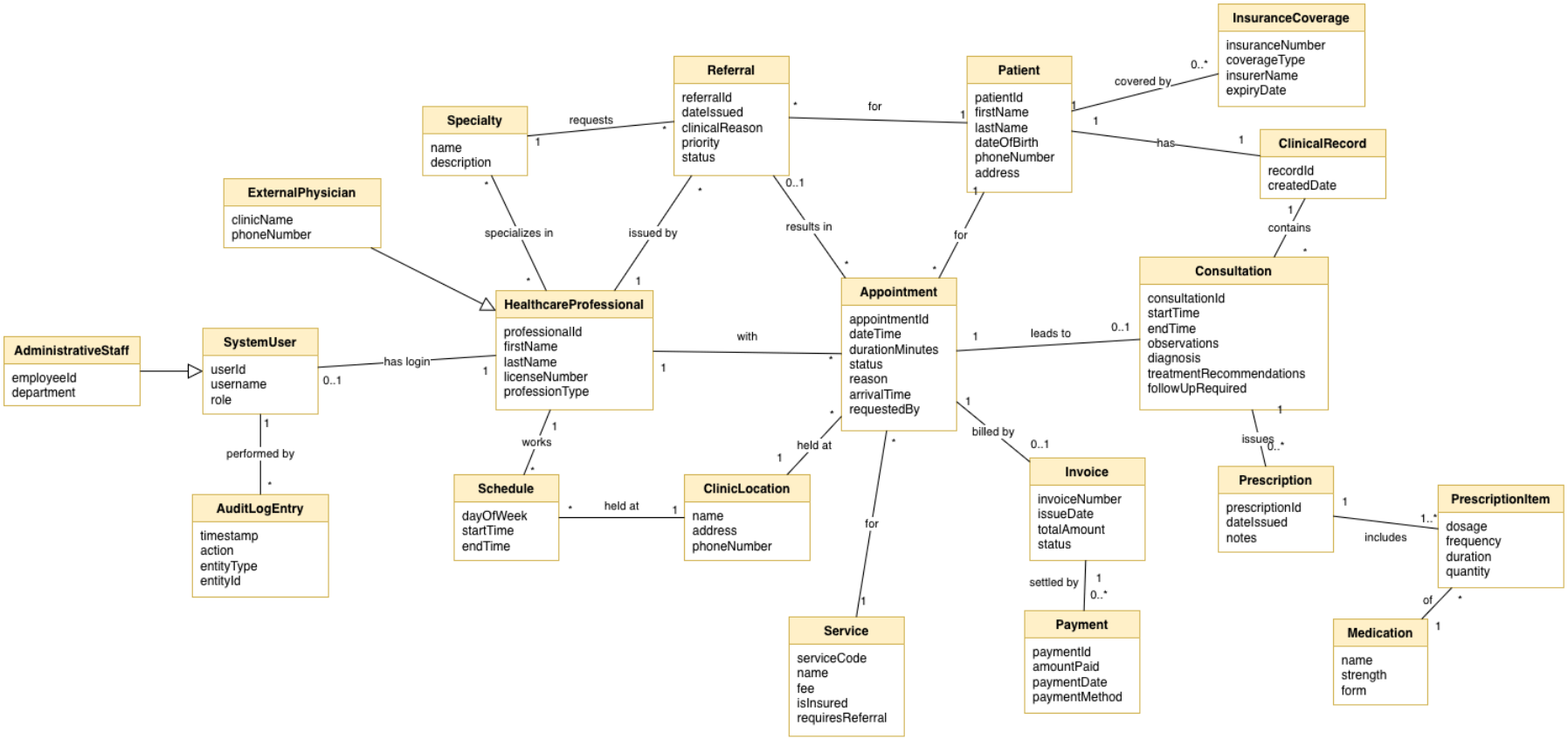
Use case classification and description:

1. Book walk in appointment
    - Classification: Critical
    - Primary actors: Patient, Administrative Staff
    - Description: The main business workflow which coordinates healthcare availability. A patient requests a visit, staff checks which doctors are working and whether the timing and date work. It is marked critical because scheduling visits is the primary function of the clinic.
2. Register patient arrival
    - Classification: Critical
    - Primary actors: Administrative staff
    - Description: When a patient walks into the clinic for the corresponding appointment, administrative staff uses this feature to check them in. The staff confirms who the patient is and checks their details. It is critical because it controls the daily patient queue and allows the staff to prevent missed or lost appointments.
3. Conduct Consultation and record notes
    - Classification: Critical
    - Actors: Healthcare Professional
    - Description: During the actual medical exam, the healthcare professional (doctor, etc.) will have their session with the patient, take corresponding notes and make a diagnosis. It allows the doctor to outline the next steps and to make the next decision. This is critical because it is necessary to create accurate medical records that reflect the patient's issues.
4. Issue specialist referral:
    - Classification: Critical
    - Actors: Family physician, external physician
    - Description: When a patient needs help that a regular doctor cannot provide, the doctor

creates a referral to send them to a specialist that can further help them with their symptoms. It is critical because specialists can only be referred to by doctors after they meet with the patient.

5. Issue prescription:
    - Classification: Critical
    - Actors: Healthcare Professional
    - Description: If a patient requires medicine, the healthcare provider uses this to write them a prescription. It records the exact medicine name and its info. It is critical because prescribing medication can only be done by doctors, and it requires careful consideration.
6. Search patient record
    - Classification: Non critical
    - Actors: Administrative staff
    - Description: This is a lookup tool used by the staff to quickly find the existing patients' info (health card number, DOB, name, etc). It is non-critical since it is a helper function that runs with other tasks like booking an appointment or checking someone in.
7. Validate medical referral
    - Classification: non critical
    - Actors: Administrative staff, patient
    - Description: This feature checks whether a referral file is real and active. It is classified as non critical since it isn't done all the time but only when a referral is given.
8. Cancel or reschedule appointment:
    - Classification: non critical
    - Actors: patient, administrative staff
    - Description: This allows a patient or a staff member to cancel or reschedule an appointment. When the appointment is cancelled or shifted time slots, the system frees up the time slot so another patient can possibly take it. It is non critical because it is considered maintenance rather than patient care.
9. Manage schedules and location
    - Classification: non critical
    - Actors: administrative staff
    - Description: Staff uses this tool to set up the doctors' timings and other logistic information, from assigning doctors to specific rooms all the way to the clinic opening times. It is non-critical, as it is like backend setup work that happens before the patients come for their appointments.
10. Process payment and invoicing
    - Classification: non critical
    - Actors: administrative staff, payment system
    - Description: When a patient receives a service that is not covered by health insurance, this feature calculates the fee and prints a receipt. It is non critical because most standard clinic visits are covered under public health insurance, not requiring payment.

II. UML Domain Model



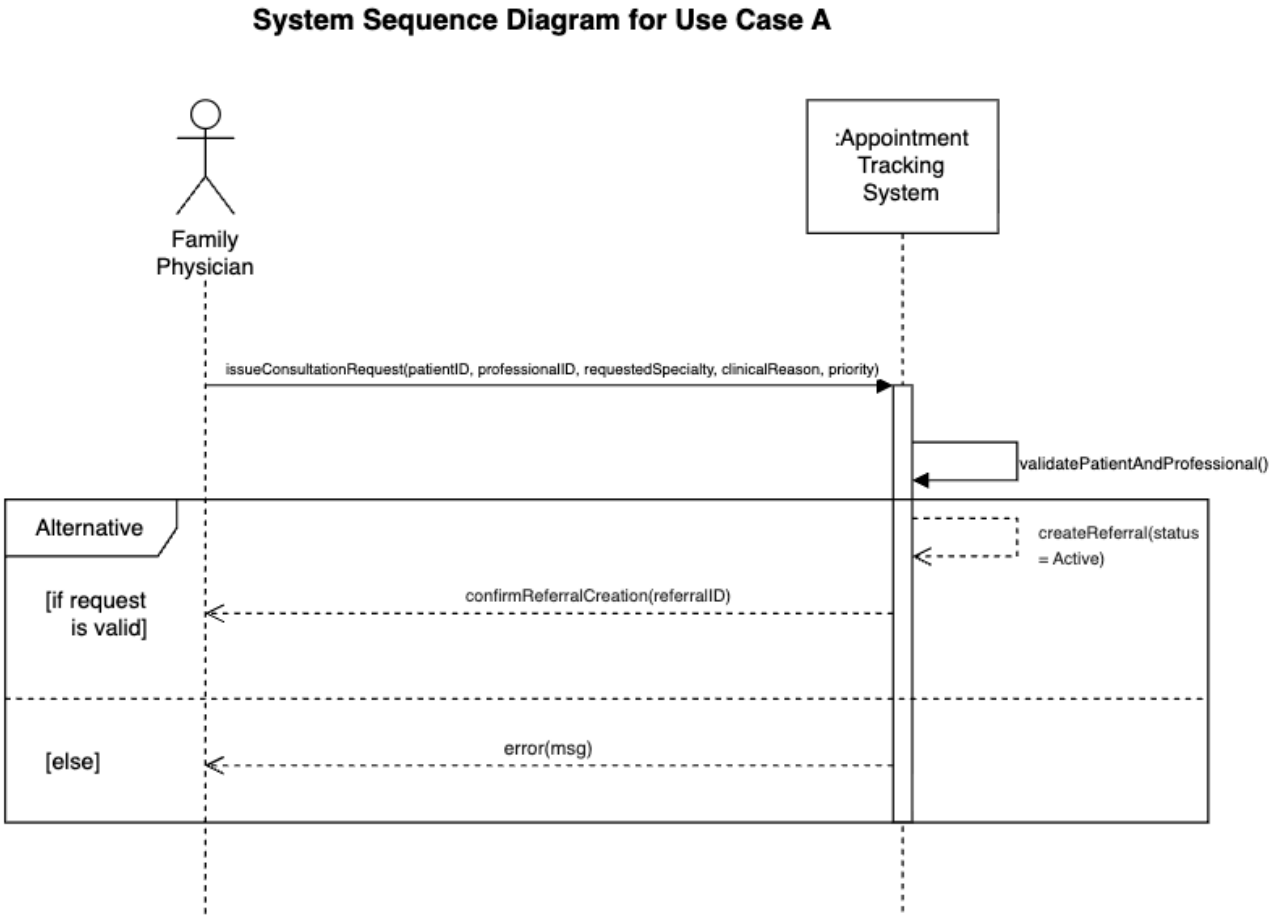
III. Three fully dressed scenarios for three critical use cases

A. A family physician may issue a consultation request for a patient to see a specialist.
    1. Success Case: The family physician receives a confirmation message that shows that the issued request has been validated, meaning that the referral status is now set to ‘active’.
    2. Failure Case: Family physician issues a consultation request with invalid (e.g. invalid patient or physician ID) or missing information (e.g. clinical reason has not been specified), prompting the system to invalidate their request and show them an error message.

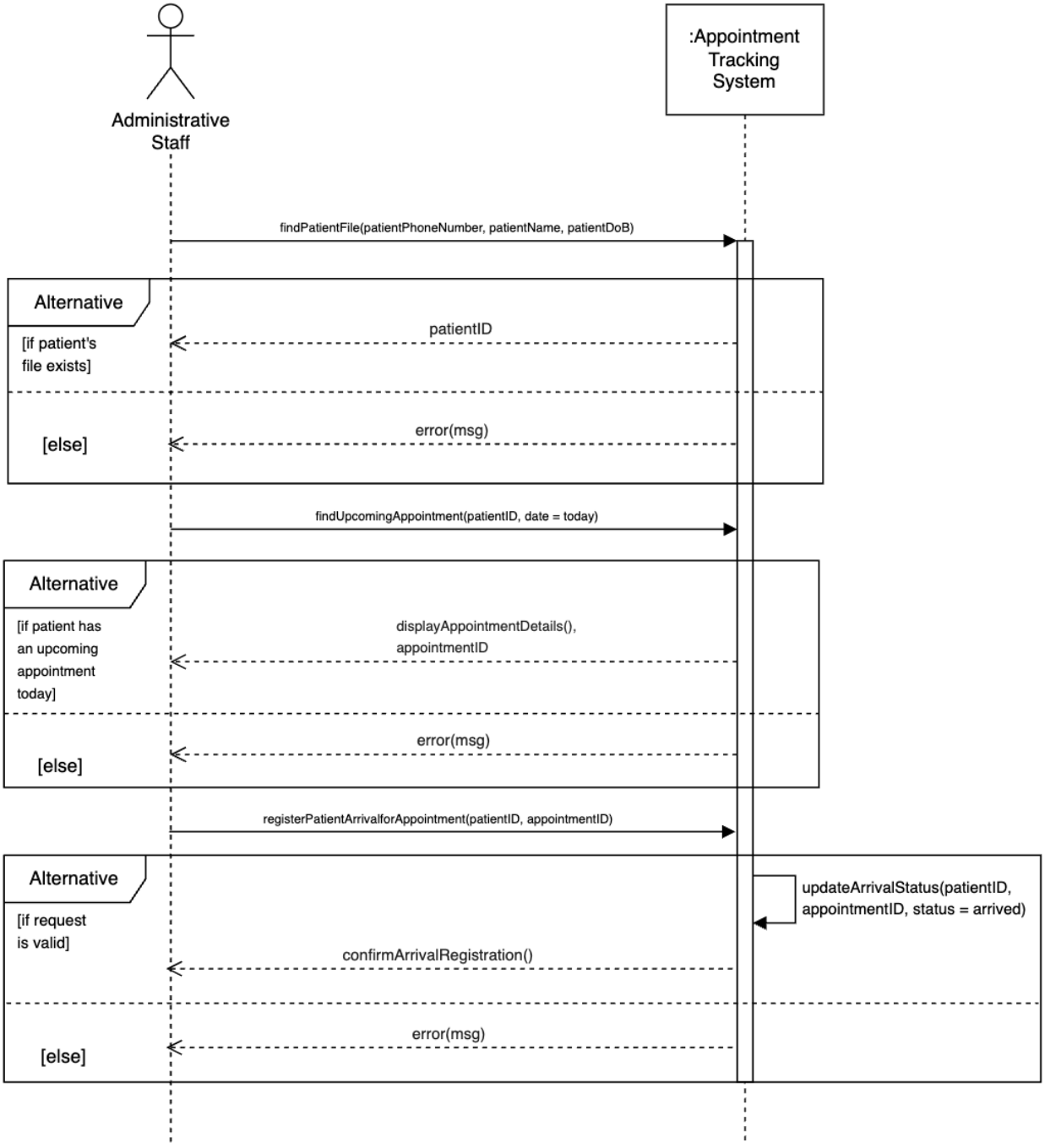
B. Administrative staff may register a patient’s arrival on the day of their appointment.
    1. Success Case: An upcoming appointment under today’s date is shown on the patient’s file, allowing the administrative staff to register the patient’s arrival.
    2. Failure Case: The patient’s file does not show any upcoming appointments today.

C. Administrative staff may book an appointment for a patient that has arrived in a clinic.
    1. Success Case: A specific kind of appointment with a designated professional is successfully booked at the desired date and time that was requested. The system also automatically calculates the appointment duration in accordance to the appointment type that was selected.
    2. Failure Case: The booking request was either submitted with invalid information or the professional/timeslot was not available to be booked.

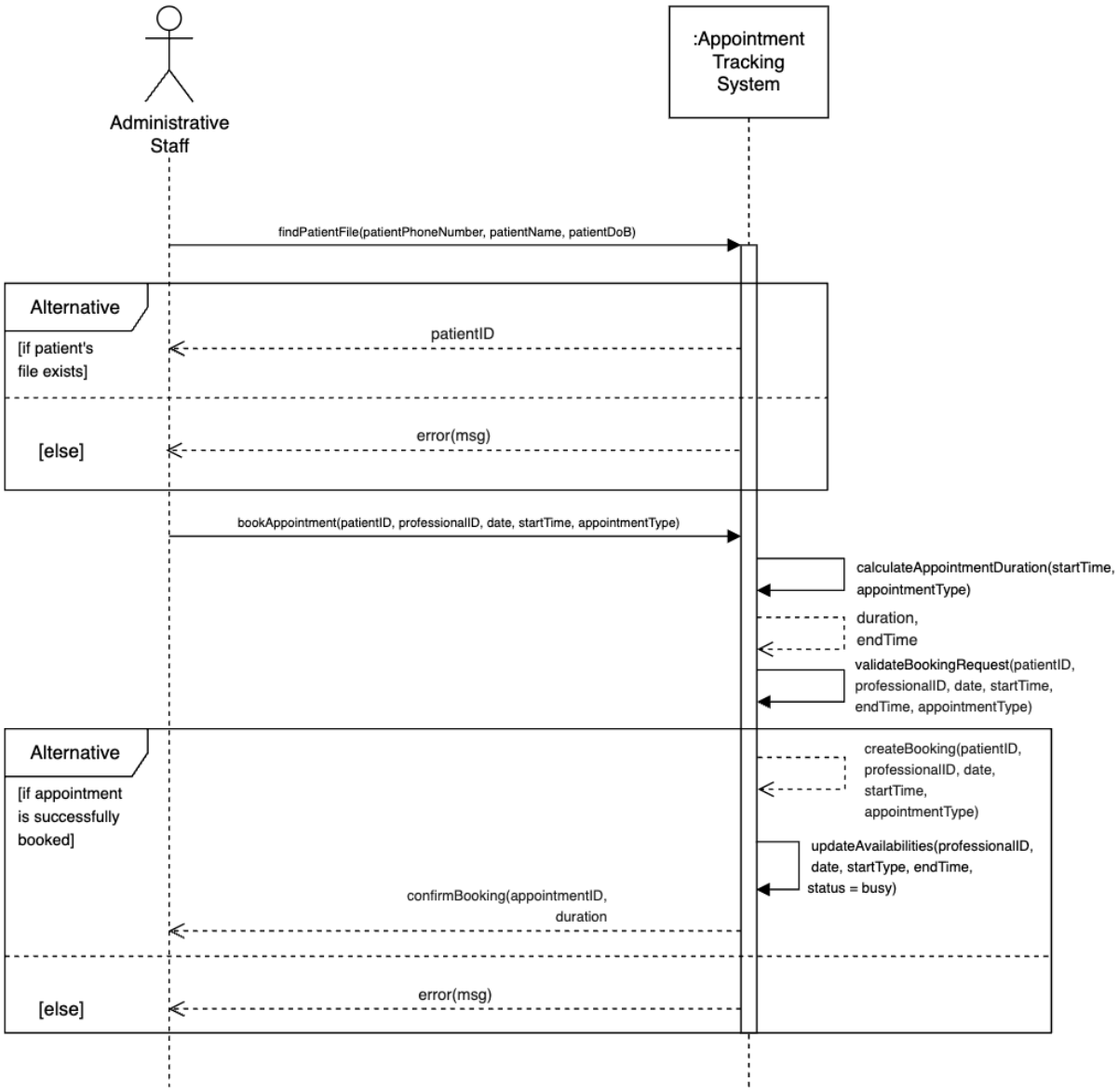
IV. System Sequence Diagrams



System Sequence Diagram for Use Case B



System Sequence Diagram for Use Case C



V. Identification of System Operations and Operation Contracts.

Contract CO1: submitConsultationRequest

Operation	submitConsultationRequest()
Cross reference	Use Case: Issue a Consultation Request (UC1)
Pre-conditions	- There is a consultation request underway: the patient, the requested specialty and/or requested professional, the clinical reason and the priority have been entered (*startConsultationRequest*, *specifyRecipient*, *enterReferralDetails*). - The user is logged in as a family physician (A1). - The patient, the requested specialty and/or professional exist, and the clinical reason and priority are not empty.
Post-conditions	1. Instance creation and deletion a. A Referral instance *r* was created. 2. Attribute modification a. *r.clinicalReason* was set to *clinicalReason* and *r.priority* was set to priority (attribute modification). b. *r.dateIssued* was set to the current date (attribute modification). c. *r.status* was set to "*active*" (attribute modification). 3. Associations formed or broken a. *r* was associated with the Patient (association formed). b. *r* was associated with the issuing HealthcareProfessional (association formed). c. *r* was associated with the requested Specialty and/or HealthcareProfessional (association formed).

Contract CO2: registerArrival

Operation	registerArrival(appointmentId: AppointmentID)
Cross reference	Use Case: Register a Patient's Arrival (UC2).
Pre-conditions	- A patient has been selected, and an Appointment with the given appointmentId exists for that patient. - *a.dateTime* is today's date and a.status is "confirmed". - The user is logged in as administrative staff (A1).
Post-conditions	1. Instance creation and deletion No instances were created or deleted. 2. Attribute modification a. *a.arrivalTime* was set to the current time (attribute modification). b. *a.status* was set to "arrived" (attribute modification). 3. Associations formed or broken No associations were formed or broken.

Contract CO3: confirmAppointment

Operation	confirmAppointment(professionalId: ProfessionalID, clinicId: ClinicID, dateTime: DateTime, durationMinutes: integer, reason: String [optional])
Cross reference	Use Case: Book an Appointment (UC3)
Pre-conditions	- There is an appointment booking underway: a *Patient* and a *Service* have been selected. - If *Service.requiresReferral* is true, a selected Referral *r* belongs to the same patient, has *r.status* “active”, and matches the chosen professional (requested professional, or requested specialty among the professional’s specialties). - The *HealthcareProfessional* has a *Schedule* entry at the chosen *ClinicLocation* covering the requested interval, and has no overlapping “confirmed” or “arrived” *Appointment*. This is checked at confirmation time. - The user is logged in as administrative staff (A1).
Post-conditions	1. Instance creation and deletion a. An Appointment instance *a* was created. 2. Attribute modification a. *a.dateTime* was set to *dateTime*, *a.durationMinutes* to *durationMinutes*, and *a.reason* to reason, if given (attribute modification). b. *a.status* was set to “confirmed” (attribute modification). 3. Associations formed or broken a. *a* was associated with the *Patient*, the *HealthcareProfessional*, the *ClinicLocation* and the *Service*, based on *id* match (association formed). b. *a* was associated with the Referral *r*, if one was used (association formed).